

Sai Srinivas Pedhapolla

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PROFESSIONAL SUMMARY

- Data Engineer with 2+ years of experience designing, building, and supporting scalable ETL/ELT pipelines, cloud data platforms, and analytics-ready datasets across AWS and GCP.
- Proficient in Python, SQL, PySpark, Databricks, Snowflake, data warehousing, multi-source ingestion, schema design, transformation, validation, orchestration, monitoring, and CI/CD.
- Experienced in platform migrations, workflow automation, production troubleshooting, data quality controls, and delivery of reliable datasets for reports, dashboards, APIs, and downstream applications.
- Combines data engineering expertise with Angular and TypeScript front-end development to deliver end-to-end solutions from ingestion and storage through REST APIs and user-facing interfaces.

TECHNICAL SKILLS

Languages & Data Processing: Python, SQL, PL/SQL, Spark SQL, PySpark, Pandas, DuckDB, Parquet

Cloud & Infrastructure: AWS (S3, Glue, Lambda, Redshift, ECR, ECS Fargate, EventBridge, CloudWatch, IAM), GCP (Cloud Functions, Cloud Scheduler, Pub/Sub, Cloud Monitoring), Databricks, Docker, GitHub Actions CI/CD

Data Architecture: ETL/ELT design, schema & metadata design, data warehousing, source-to-target integrity, Snowflake, Firebase

APIs & Automation: REST API integration, JSON/XML/CSV ingestion, event-driven automation, workflow orchestration

Frontend & Web: Angular, TypeScript, JavaScript, HTML5, CSS3, REST API integration, responsive UI development, component-based architecture

Analytics & BI: Google Analytics 4 (GA4) API, Streamlit, Plotly, Power BI, Excel, dashboard-ready data prep

Quality & Practice: data validation, pipeline monitoring, root cause analysis, Agile, code review, documentation

PROFESSIONAL EXPERIENCE

Data Engineering / Analyst

Goods Unite Us

Sep 2025 – Present

Wisconsin, United States

- Built and maintained Python, PySpark, SQL, ETL workflows to ingest, transform, and validate structured and semi-structured data from multiple sources for analytics and downstream applications.
- Led schema-aware data modeling for a Snowflake-to-Firebase migration, redesigning relational structures into document collections to improve query performance and reduce infrastructure cost.
- Configured GCP monitoring and alerting (log-based metrics, notification channels) to surface pipeline failures and data-freshness breaches, cutting time-to-detection on production issues.
- Automated recurring jobs with GCP Cloud Scheduler, Cloud Functions, and Pub/Sub, replacing manual runs with scheduled, event-driven execution and automated failure alerts.
- Containerized ETL jobs with Docker and deployed to AWS ECS Fargate with EventBridge scheduling, IAM-scoped access, CloudWatch logging, and GitHub Actions CI/CD for repeatable releases.
- Built and maintained Angular and TypeScript front-end components that consumed internal REST APIs, giving stakeholders self-serve access to pipeline outputs and reducing ad-hoc data requests.
- Implemented source-to-target validation, schema checks, logging, and exception handling across ingestion workflows to improve data quality, traceability, and operational support for production pipelines.
- Collaborated with analysts, developers, and business stakeholders to translate reporting and application requirements into reusable data models, documented workflows, and maintainable engineering solutions.
- Supported deployment readiness through code reviews, environment configuration, operational documentation, and post-release monitoring of cloud-based data services.

Python Developer & R&D Data Intern

Webdaddy

Aug 2024 – Feb 2025

United States

- Developed Python, SQL, and Pandas-based ETL workflows to ingest and transform REST API and multi-source data for analytics and reporting platforms.
- Integrated XML, JSON, and CSV inputs into a unified transformation layer, improving pipeline flexibility for semi-structured and unstructured data.
- Investigated pipeline and performance issues through root cause analysis, implementing fixes that improved execution stability and reduced recurring failures.
- Partnered with stakeholders in an Agile environment to align technical delivery with evolving business requirements.
- Created reusable transformation modules, technical documentation, and testable data-processing components to support consistent releases, code reviews, and future pipeline enhancements.
- Applied data profiling and transformation checks to identify malformed records, schema inconsistencies, and missing values before datasets reached analytics users.

R&D Data Analyst Intern

May 2022 – Dec 2023

Findem, Inc.

India

- Performed business data analysis on large datasets using SQL to surface trends and recommendations supporting KPI tracking and decision-making.
- Prepared and validated datasets to improve data quality and downstream usability for reporting and analytics workflows.
- Built dashboard-ready outputs using SQL, Excel, and Power BI, improving business visibility into key metrics.
- Resolved data and reporting issues through validation checks and root cause analysis, strengthening data accuracy and stakeholder trust.
- Documented data definitions, transformation logic, and reporting assumptions to improve metric consistency and support communication between technical and non-technical stakeholders.
- Translated business questions into SQL analyses, repeatable reporting logic, and clearly communicated findings for operational and product decision-making.

PROJECTS

AWS-Based FEC Political Donation Data Pipeline

- Built Python ETL pipelines ingesting daily FEC filing data from government-hosted S3 sources into Parquet, with incremental + backfill logic; deployed via Docker on ECS Fargate with EventBridge, CloudWatch, and GitHub Actions CI/CD.
- Designed the workflow for idempotent processing, partitioned storage, historical backfills, and failure recovery; added IAM-scoped access, centralized logging, and scheduled execution for production-ready operation.

GA4 Analytics Dashboard

- Built a Streamlit dashboard using the Google Analytics Data API, Plotly, and Pandas for real-time GA4 metrics without navigating the GA4 UI; resolved revenue reconciliation discrepancies across purchases, renewals, and refunds.
- Modeled API responses into analysis-ready tables, applied validation for purchases, renewals, and refunds, and developed interactive visualizations to support faster investigation of traffic, engagement, and revenue trends.

Snowflake to Firebase Data Migration

- Analyzed Snowflake schemas and dataset dependencies to design a migration strategy that reduced infrastructure cost while preserving application data accessibility.
- Mapped relational entities and dependencies to Firebase collections, defined migration and validation steps, and considered query access patterns, referential integrity, and rollback requirements during the target design.

EDUCATION

Master of Science in Data Science – New Jersey Institute of Technology, Newark

Bachelor of Technology in Computer Science and Engineering – SCSVMV University, Kanchipuram